

Visión por Computador

Versión 2026

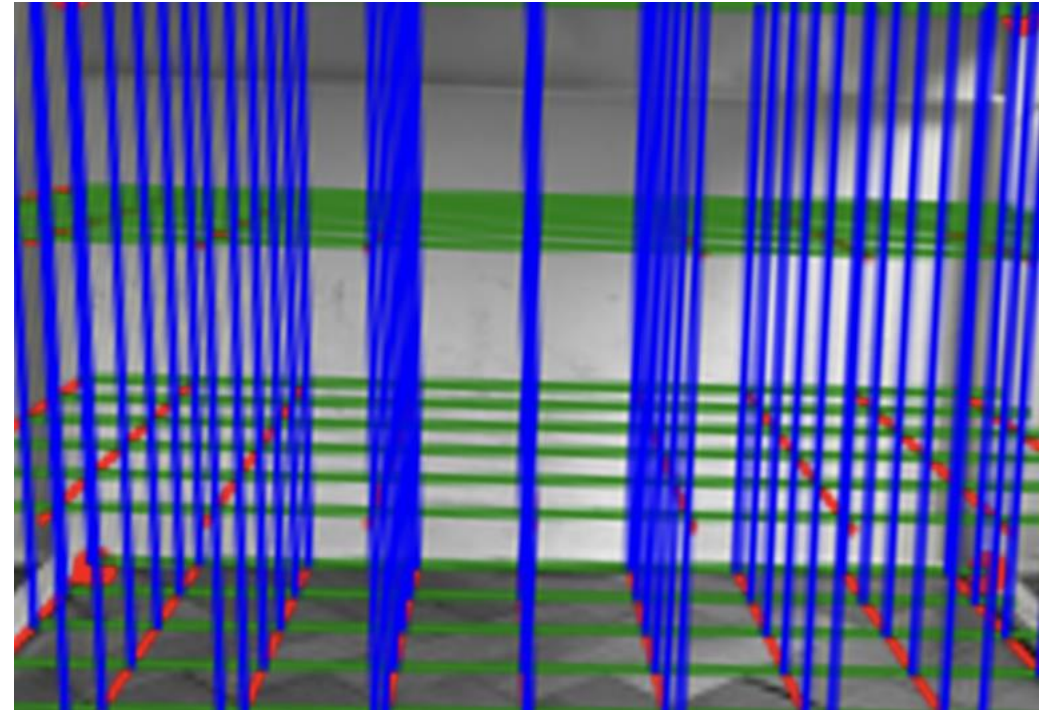


Calibración

(CAPÍTULO 2 – CV02D)

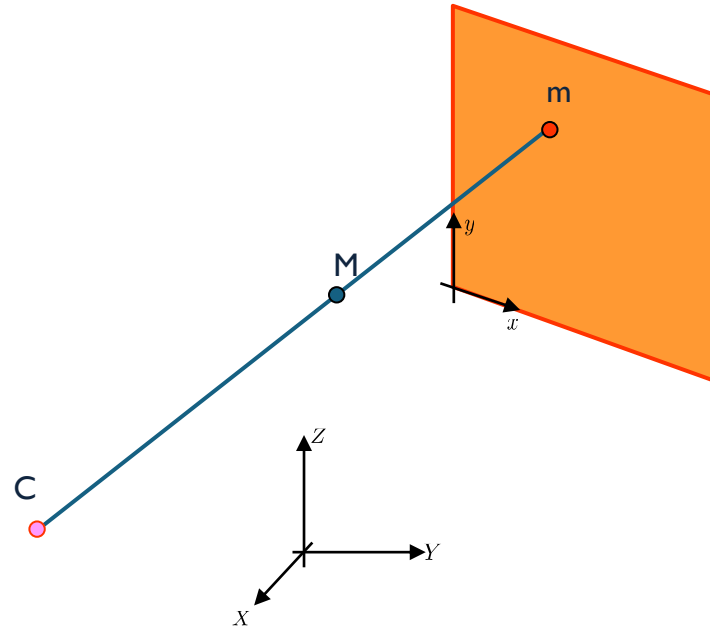
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[Calibration]

Calibration: Estimation of the parameters of the projection model



$$\lambda \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

$$\lambda \mathbf{m} = \mathbf{P} \mathbf{M}$$

$[3 \times 1] \quad [3 \times 4] [4 \times 1]$

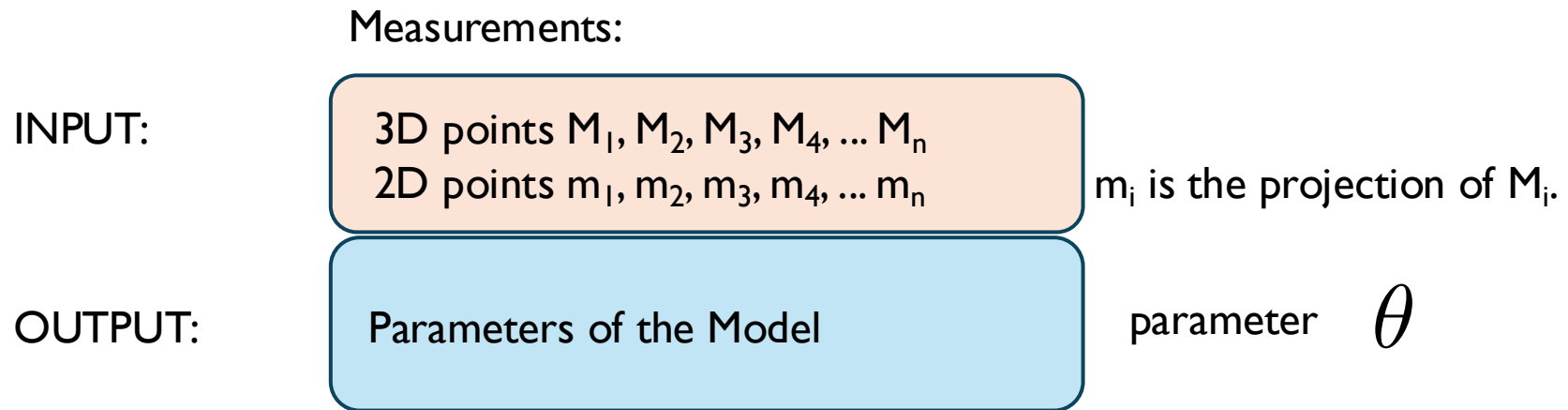
[Calibration]

Calibration is the process that estimates the parameters of the camera given corresponding points 3D – 2D

INPUT:	3D points $M_1, M_2, M_3, M_4, \dots M_n$ 2D points $m_1, m_2, m_3, m_4, \dots m_n$	m_i is the projection of M_i .
OUTPUT:	Parameters of the Model	

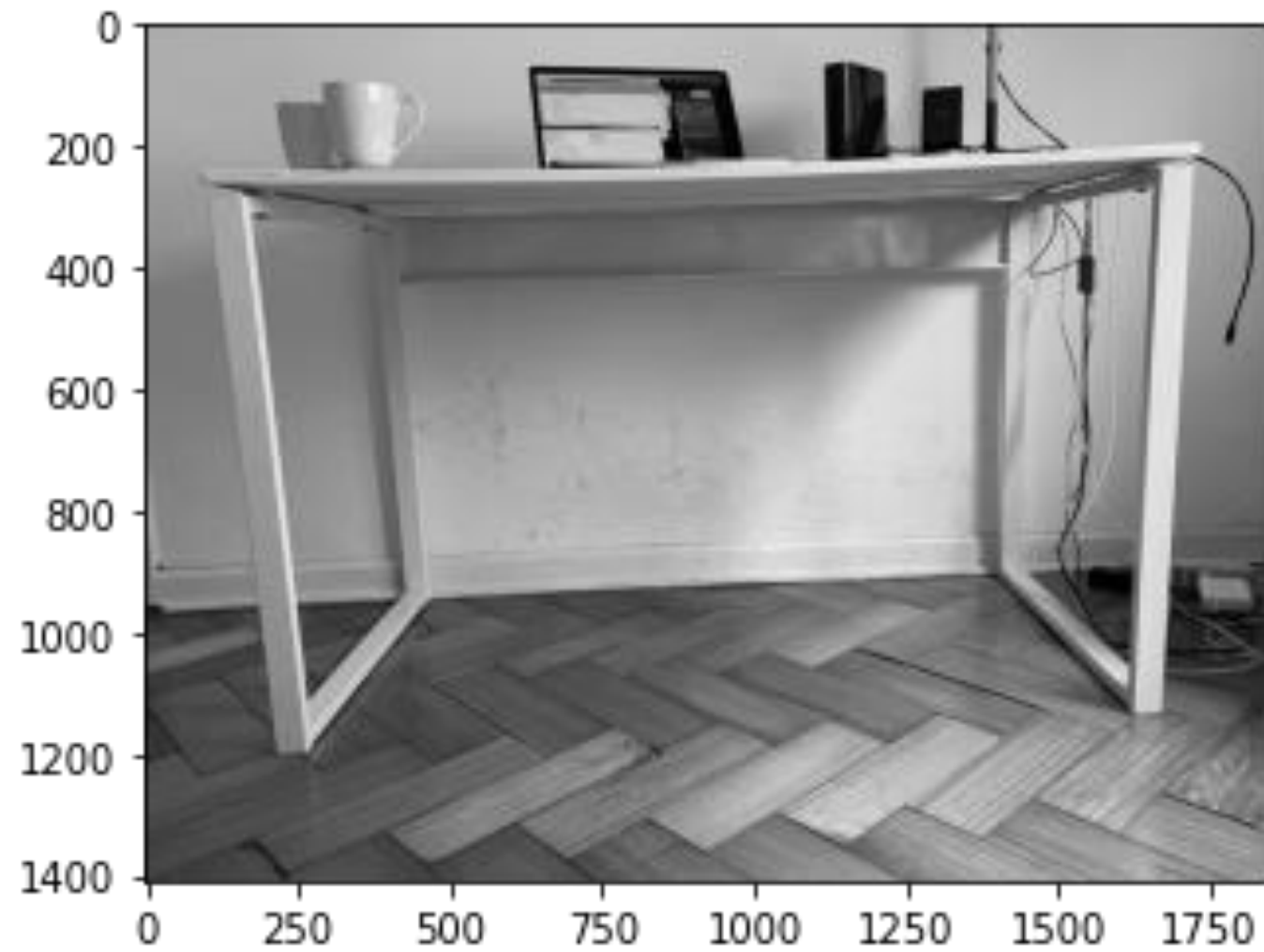
[Calibration]

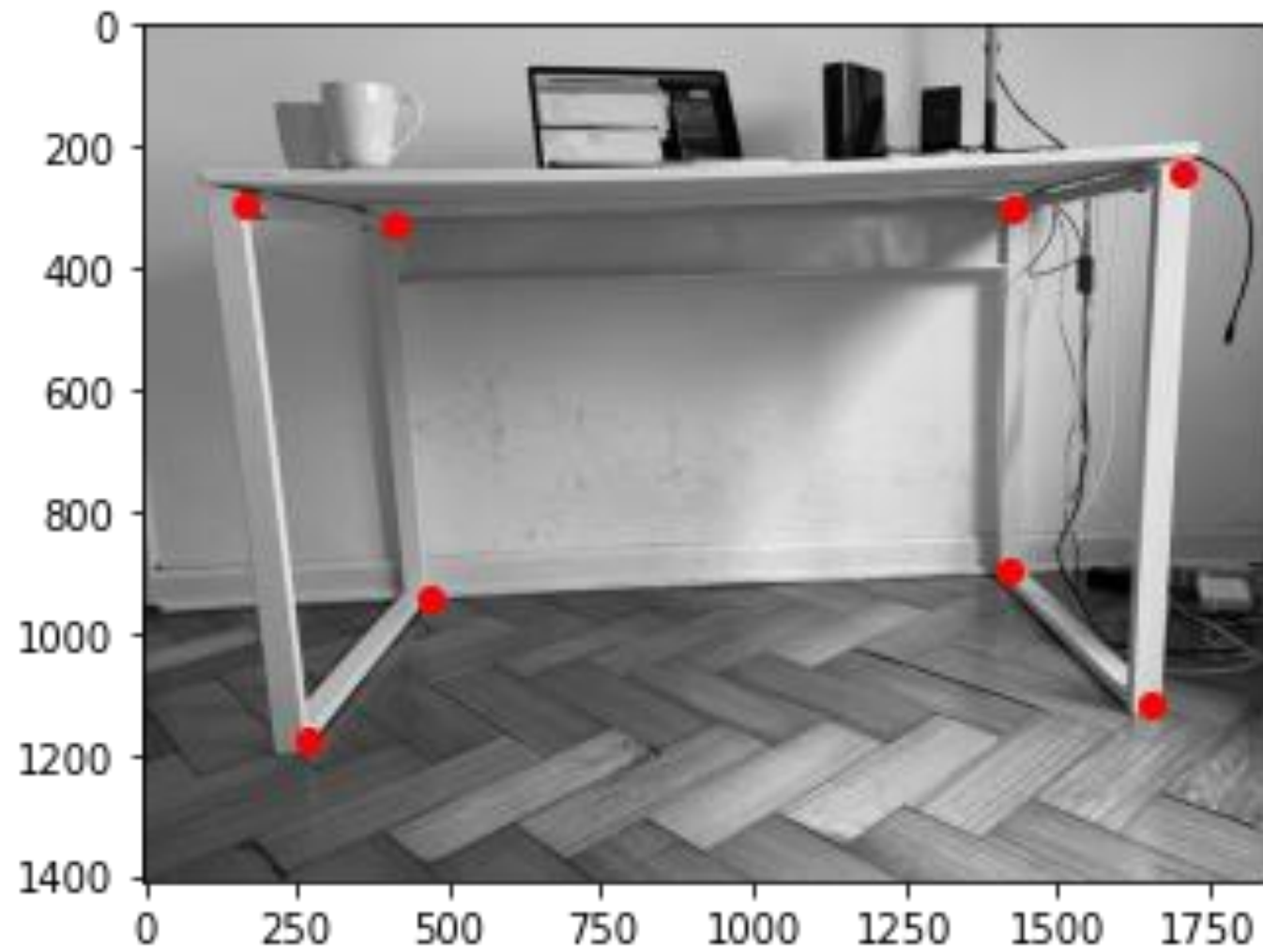
Calibration is the process that estimates the parameters of the camera given corresponding points 3D – 2D



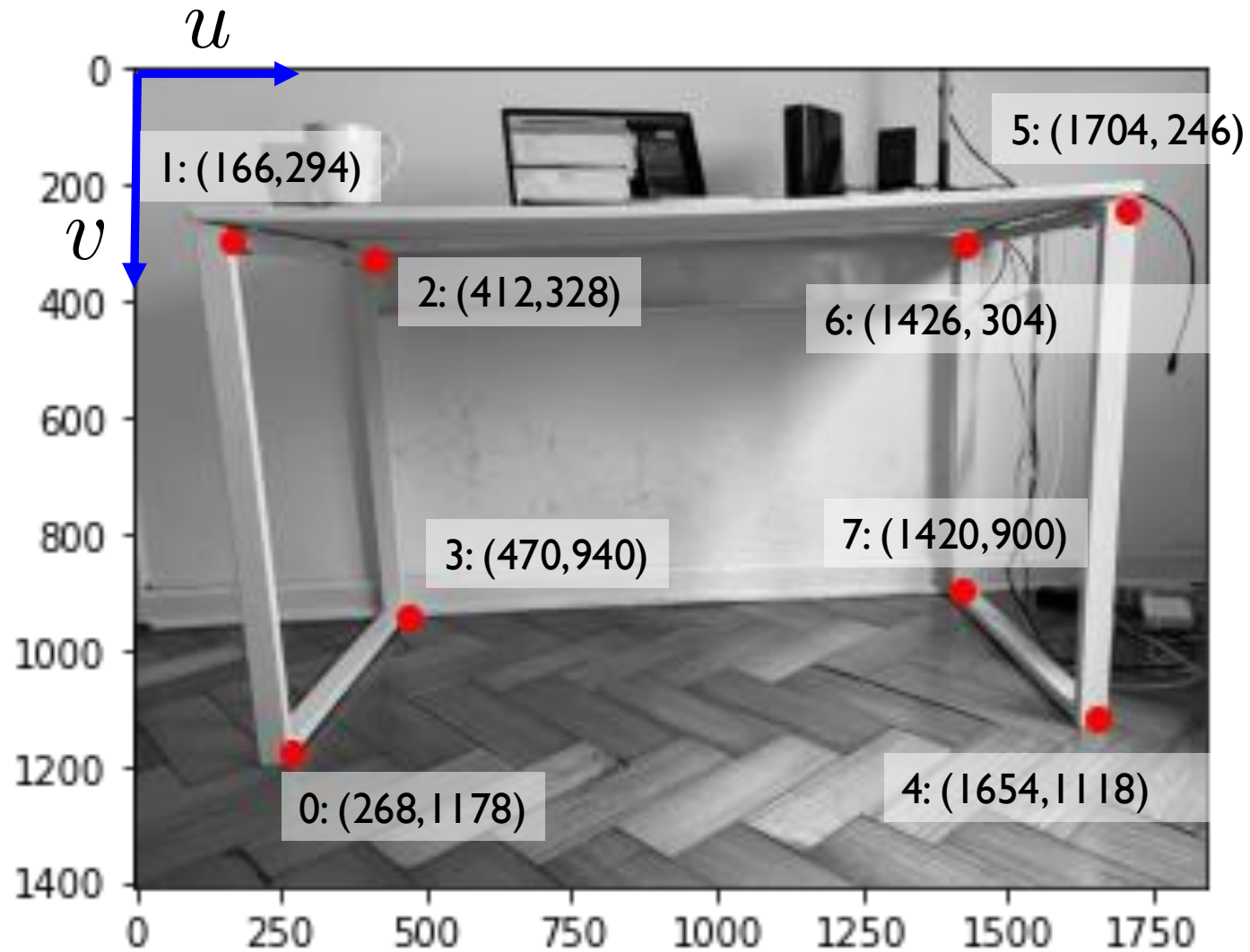
$$\hat{\mathbf{m}}_i = \mathbf{F}(\mathbf{M}_i, \theta) \quad \text{model}$$
$$J(\theta) = \frac{1}{n} \sum_{i=1}^n ||\mathbf{m}_i - \hat{\mathbf{m}}_i||$$

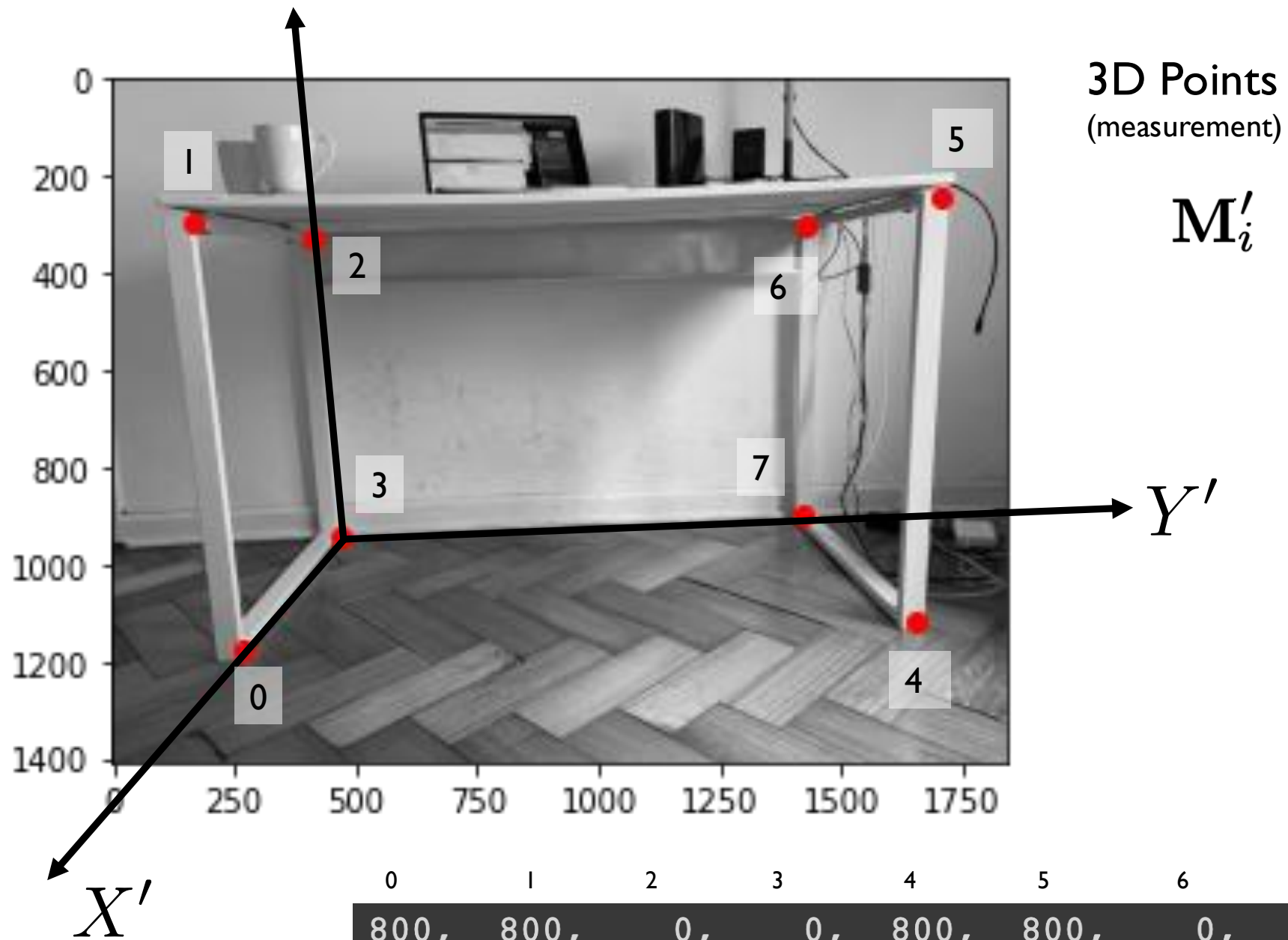
[Simple Calibration]





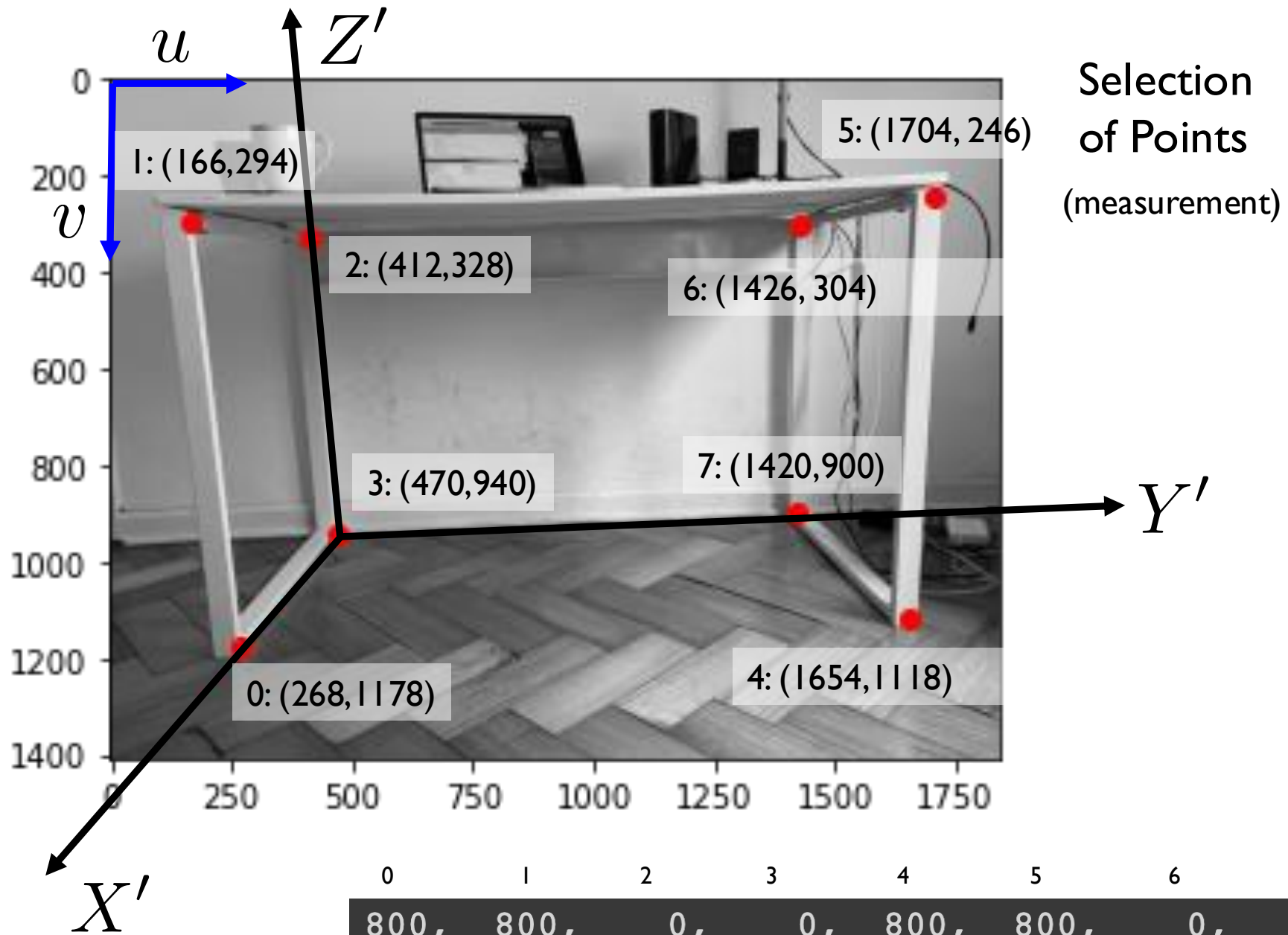
Selection
of Points





3D POINTS
(in mm)

0	1	2	3	4	5	6	7
800,	800,	0,	0,	800,	800,	0,	0
0,	0,	0,	0,	1250,	1250,	1250,	1250
0,	900,	900,	0,	0,	900,	900,	0



3D POINTS
(in mm)

0	1	2	3	4	5	6	7
800,	800,	0,	0,	800,	800,	0,	0
0,	0,	0,	0,	1250,	1250,	1250,	1250
0,	900,	900,	0,	0,	900,	900,	0

For each point $(X'_i, Y'_i, Z'_i) \rightarrow (u_i, v_i)$

$$\lambda_i \begin{bmatrix} u_i \\ v_i \\ 1 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & 1 \end{bmatrix} \begin{bmatrix} X'_i \\ Y'_i \\ Z'_i \\ 1 \end{bmatrix}$$

$$\lambda_i u_i = a_{11} X'_i + a_{12} Y'_i + a_{13} Z'_i + a_{14}$$

$$\lambda_i v_i = a_{21} X'_i + a_{22} Y'_i + a_{23} Z'_i + a_{24}$$

$$\lambda_i = a_{31} X'_i + a_{32} Y'_i + a_{33} Z'_i + 1$$

$$\begin{aligned}\lambda_i u_i &= a_{11} X'_i + a_{12} Y'_i + a_{13} Z'_i + a_{14} \\ \lambda_i v_i &= a_{21} X'_i + a_{22} Y'_i + a_{23} Z'_i + a_{24} \\ \lambda_i &= a_{31} X'_i + a_{32} Y'_i + a_{33} Z'_i + 1\end{aligned}$$

$$\begin{aligned}\lambda_i u_i &= a_{11} X'_i + a_{12} Y'_i + a_{13} Z'_i + a_{14} \\ \lambda_i v_i &= a_{21} X'_i + a_{22} Y'_i + a_{23} Z'_i + a_{24} \\ \lambda_i &= a_{31} X'_i + a_{32} Y'_i + a_{33} Z'_i + 1\end{aligned}$$

$$\begin{aligned}(a_{31} X'_i + a_{32} Y'_i + a_{33} Z'_i + 1) u_i &= a_{11} X'_i + a_{12} Y'_i + a_{13} Z'_i + a_{14} \\ (a_{31} X'_i + a_{32} Y'_i + a_{33} Z'_i + 1) v_i &= a_{21} X'_i + a_{22} Y'_i + a_{23} Z'_i + a_{24}\end{aligned}$$

$$\begin{aligned}a_{11} X'_i + a_{12} Y'_i + a_{13} Z'_i + a_{14} - a_{31} X'_i u_i - a_{32} Y'_i u_i - a_{33} Z'_i u_i &= u_i \\ a_{21} X'_i + a_{22} Y'_i + a_{23} Z'_i + a_{24} - a_{31} X'_i v_i - a_{32} Y'_i v_i - a_{33} Z'_i v_i &= v_i\end{aligned}$$

$$\begin{aligned}
a_{11}X'_i + a_{12}Y'_i + a_{13}Z'_i + a_{14} - a_{31}X'_i u_i - a_{32}Y'_i u_i - a_{33}Z'_i u_i &= u_i \\
a_{21}X'_i + a_{22}Y'_i + a_{23}Z'_i + a_{24} - a_{31}X'_i v_i - a_{32}Y'_i v_i - a_{33}Z'_i v_i &= v_i
\end{aligned}$$

$$\begin{aligned}
a_{11}X'_i + a_{12}Y'_i + a_{13}Z'_i + a_{14} - a_{31}X'_iu_i - a_{32}Y'_iu_i - a_{33}Z'_iu_i &= u_i \\
a_{21}X'_i + a_{22}Y'_i + a_{23}Z'_i + a_{24} - a_{31}X'_iv_i - a_{32}Y'_iv_i - a_{33}Z'_iv_i &= v_i
\end{aligned}$$

$$\begin{bmatrix}
X'_i & 0 \\
Y'_i & 0 \\
Z'_i & 0 \\
1 & 0 \\
0 & X'_i \\
0 & Y'_i \\
0 & Z'_i \\
0 & 1 \\
-X'_iu_i & -X'_iv_i \\
-Y'_iu_i & -Y'_iv_i \\
-Z'_iu_i & -Z'_iv_i
\end{bmatrix}^T
\begin{bmatrix}
a_{11} \\
a_{12} \\
\vdots \\
a_{33}
\end{bmatrix}
= [u_i v_i]^T$$

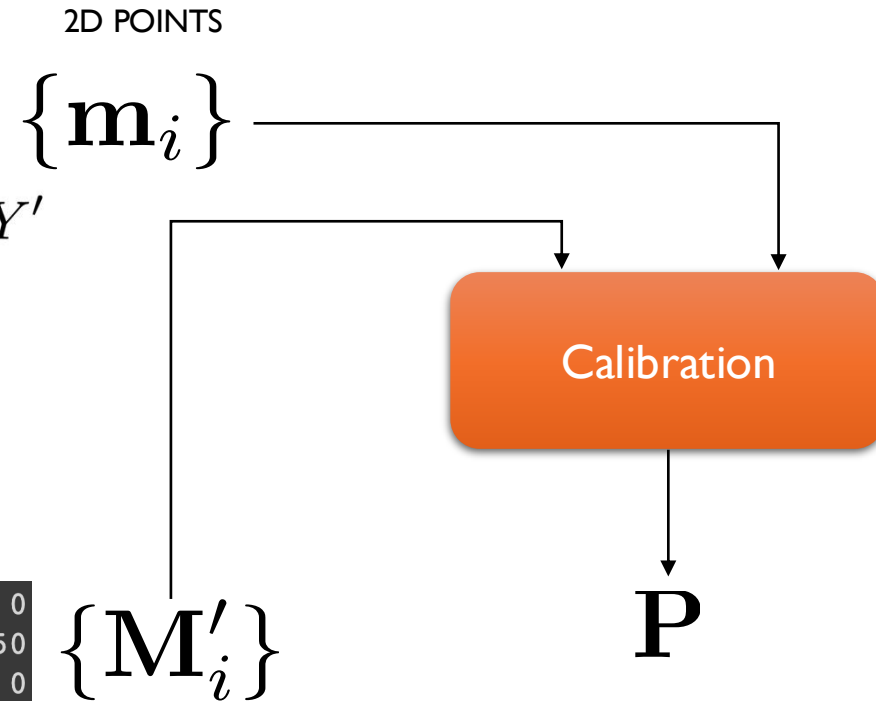
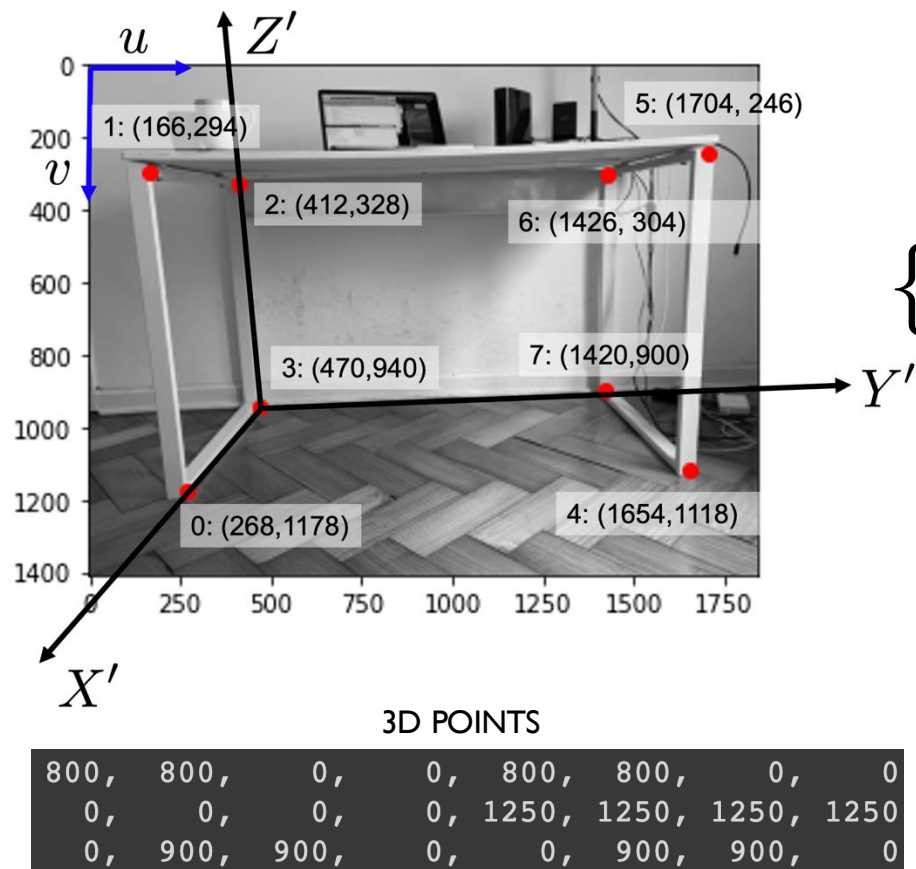
$$\begin{bmatrix} \mathbf{X}_1 \\ \mathbf{X}_1 \\ \vdots \\ \mathbf{X}_n \end{bmatrix} \mathbf{a} = \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_1 \\ \vdots \\ \mathbf{b}_n \end{bmatrix}$$

$$\mathbf{X}\mathbf{a} = \mathbf{b}$$

$$\mathbf{X}_i \mathbf{a} = \mathbf{b}_i$$

LEAST SQUARES

$$\hat{\mathbf{a}} = [\mathbf{X}^T \mathbf{X}]^{-1} \mathbf{X} \mathbf{b}$$



u = 268 166 412 470 1654 1704 1426 1420

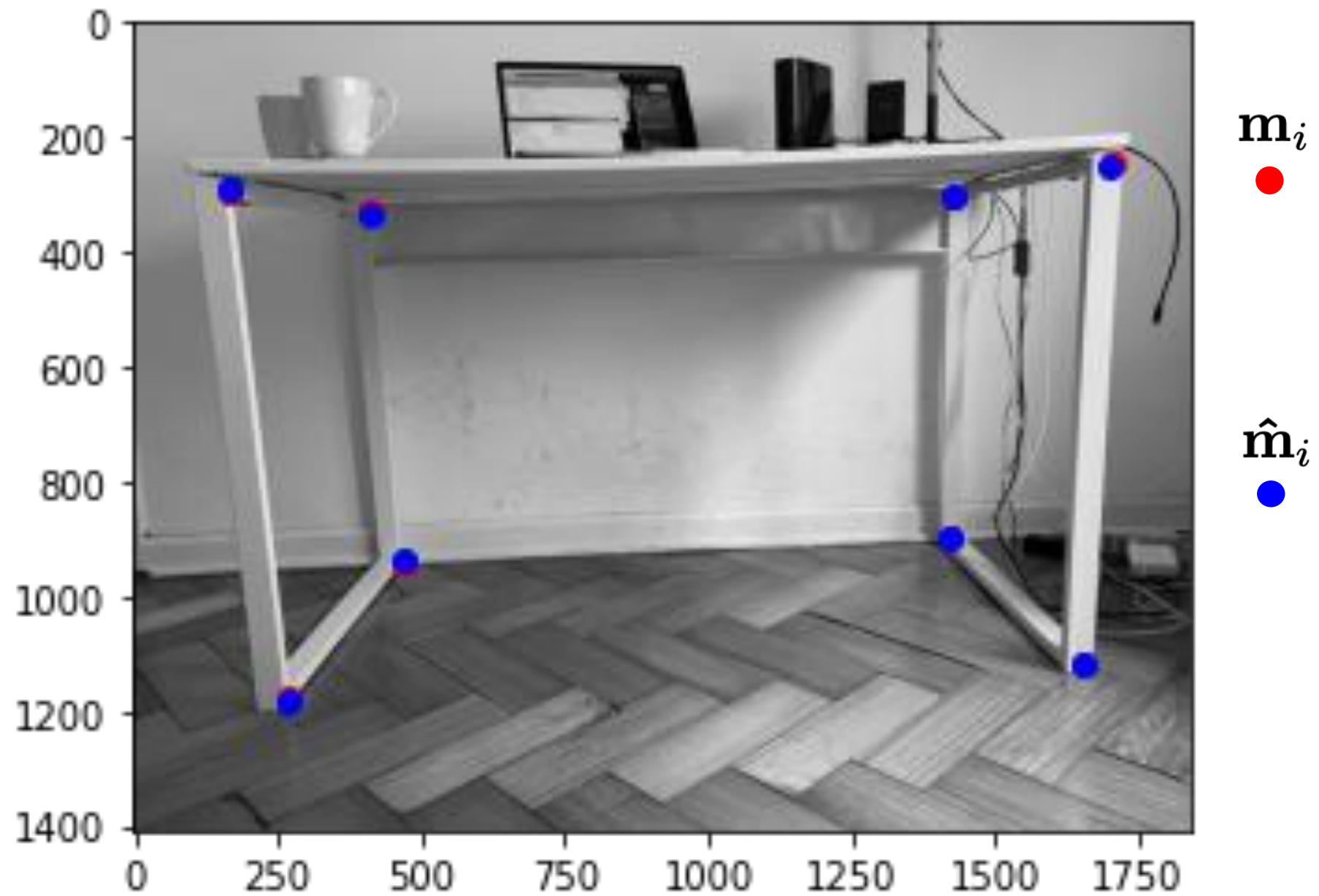
$\hat{u}$ = 271 164 412 468 1656 1699 1429 1417

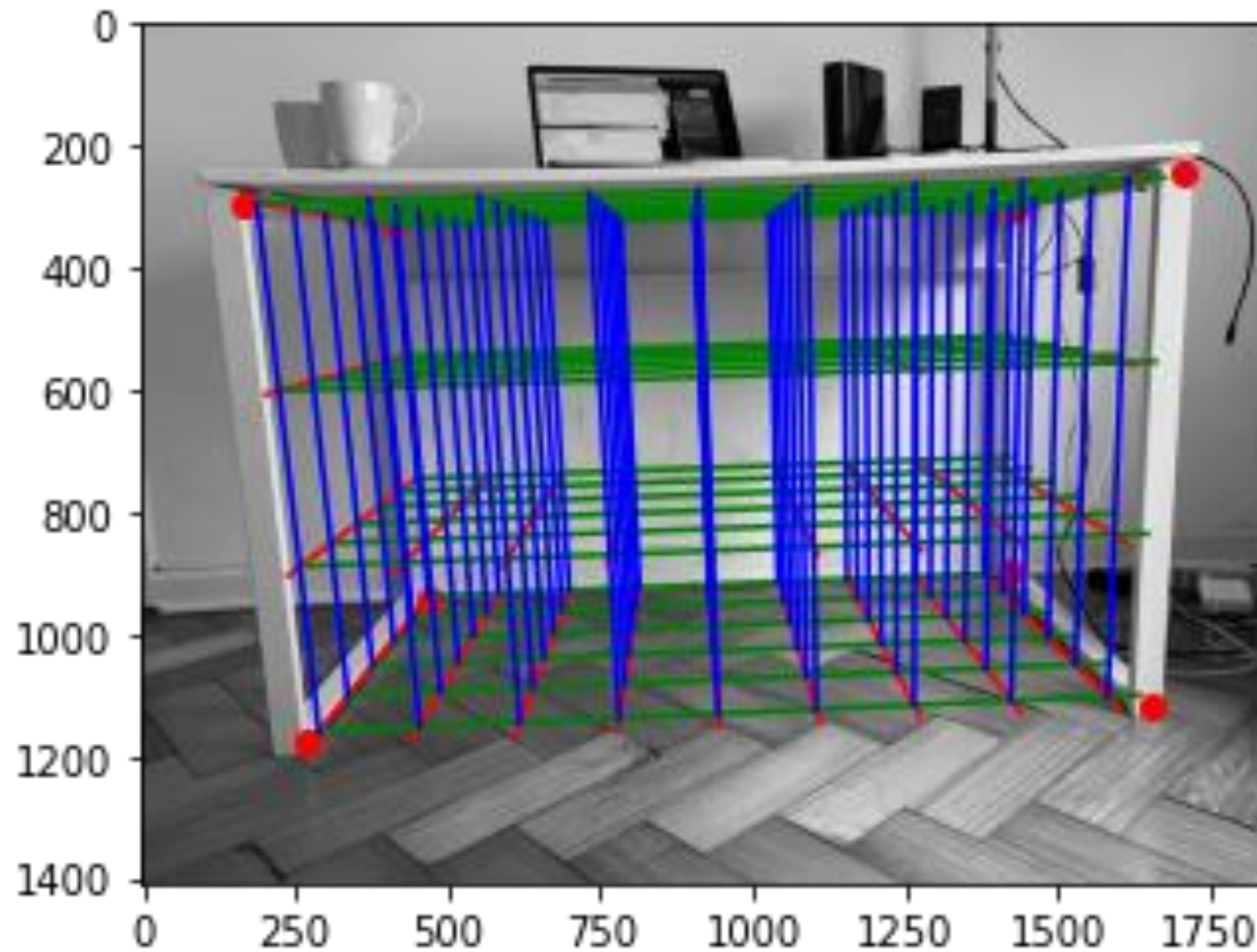
v = 1178 294 328 940 1118 246 304 900

$\hat{v}$ = 1180 289 331 937 1119 246 302 899

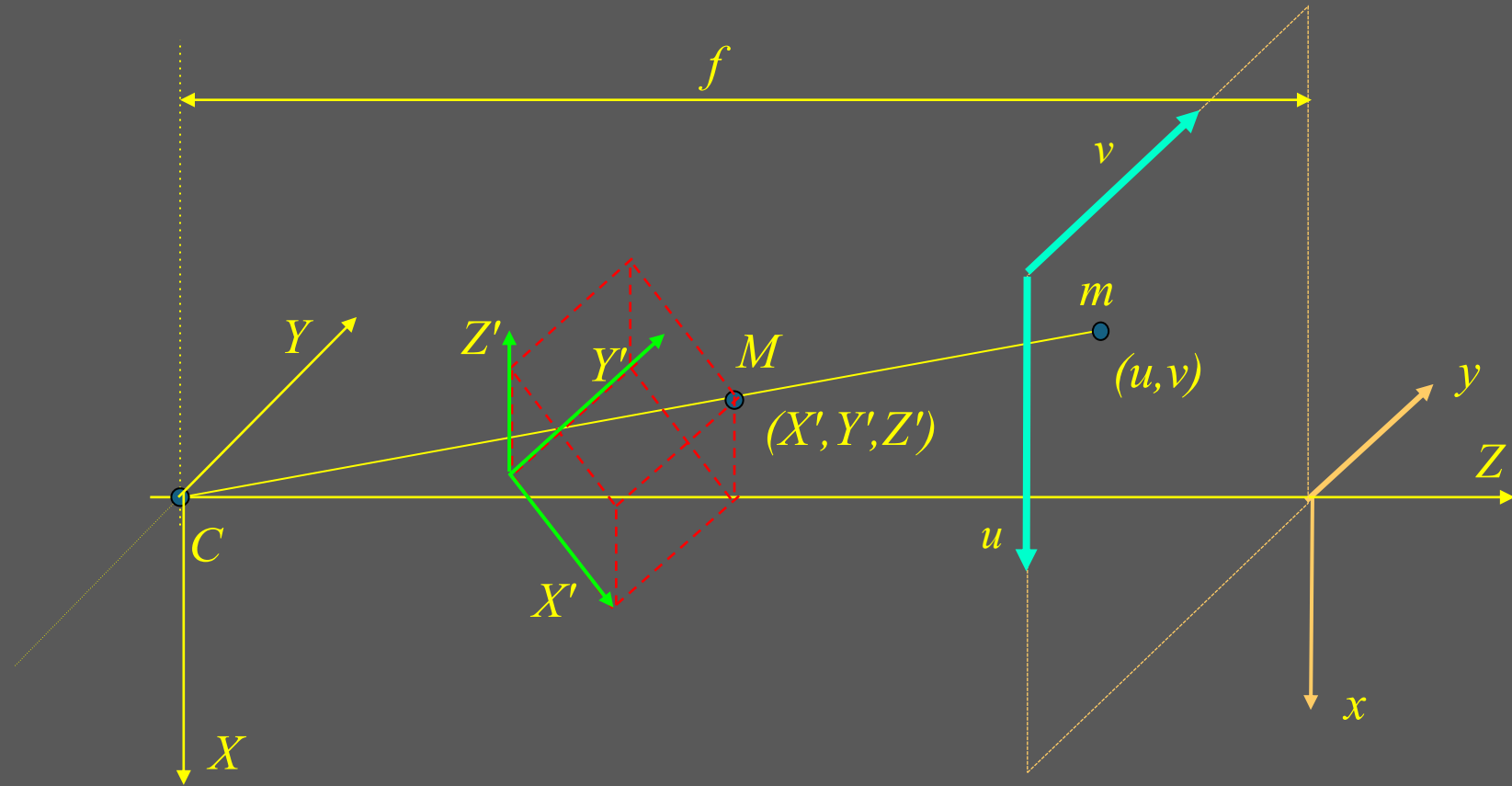
$$\lambda_i \hat{\mathbf{m}}_i = \mathbf{P} \mathbf{M}'_i$$

$$\|\hat{\mathbf{m}}_i - \mathbf{m}_i\| \rightarrow \min$$





[Non-Linear Calibration]



$$\lambda \mathbf{w} = \mathbf{KPS}'\mathbf{M}'.$$

[Camera Model]

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & s & u_0 \\ 0 & \alpha_y & v_0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} f\alpha_x & fs & u_0 & 0 \\ 0 & f\alpha_y & v_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 & 0 \\ 0 & \beta & v_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

[Camera Model]

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 & 0 \\ 0 & \beta & v_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 & 0 \\ 0 & \beta & v_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

II DoF:

- 5 intrinsec parameters $(\alpha, \beta, \gamma, u_0, v_0)$
- 6 extrinsec parameters $(\omega_X, \omega_Y, \omega_Z), (t_X, t_Y, t_Z)$

[Calibration]

Calibration is the process that estimates the parameters of the camera given corresponding points 3D – 2D

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha & \gamma & u_0 & 0 \\ 0 & \beta & v_0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}' & \mathbf{t}' \\ \mathbf{0} & 1 \end{bmatrix} \begin{bmatrix} X' \\ Y' \\ Z' \\ 1 \end{bmatrix}$$

II DoF:

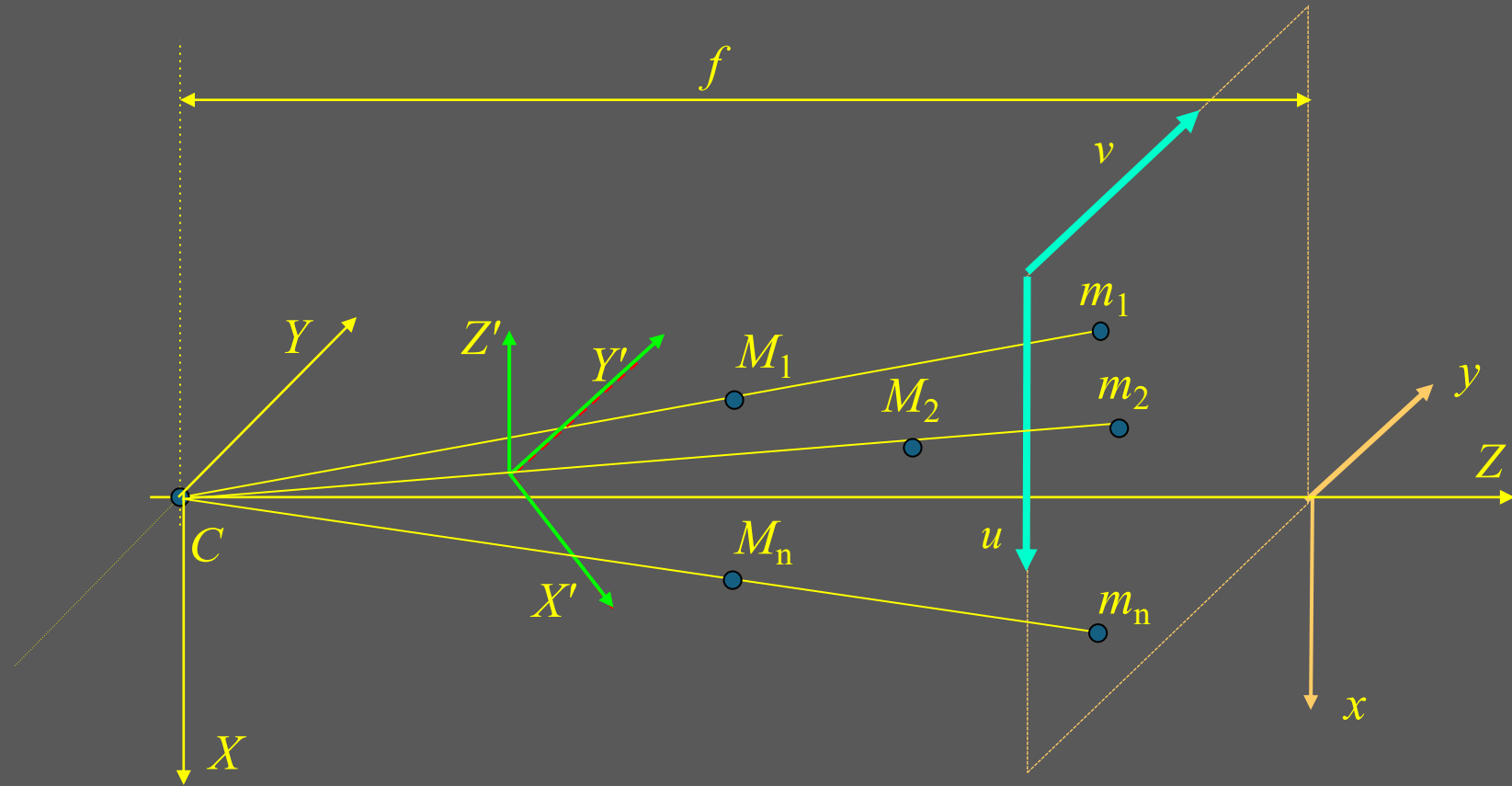
- 5 intrinsec parameters $(\alpha, \beta, \gamma, u_0, v_0)$
- 6 extrinsec parameters $(\omega_X, \omega_Y, \omega_Z), (t_X, t_Y, t_Z)$

[Calibration]

Calibration is the process that estimates the parameters of the camera given corresponding points 3D – 2D

INPUT:	3D points $M_1, M_2, M_3, M_4, \dots M_n$ 2D points $m_1, m_2, m_3, m_4, \dots m_n$	m_i is the projection of M_i .
OUTPUT:	$(\alpha, \beta, \gamma, u_0, v_0)$ $(\omega_X, \omega_Y, \omega_Z), (t_X, t_Y, t_Z)$	

[Camera Model]



[Calibration]

Calibration is the process that estimates the parameters of the camera given corresponding points 3D – 2D

Measurements:

INPUT:

3D points $M_1, M_2, M_3, M_4, \dots M_n$
2D points $m_1, m_2, m_3, m_4, \dots m_n$

m_i is the projection of M_i .

OUTPUT:

$(\alpha, \beta, \gamma, u_0, v_0)$
 $(\omega_X, \omega_Y, \omega_Z), (t_X, t_Y, t_Z)$

parameter θ

$$\hat{\mathbf{m}}_i = \mathbf{F}(\mathbf{M}_i, \theta) \quad \text{model}$$

$$J(\theta) = \frac{1}{n} \sum_{i=1}^n ||\mathbf{m}_i - \hat{\mathbf{m}}_i||$$

[Calibration]

Calibration is the process that estimates the parameters of the camera given corresponding points 3D – 2D

For each corresponding pair of points $\mathbf{m}_i - \mathbf{M}_i$:

$$\hat{\mathbf{m}}_i = \mathbf{F}(\mathbf{M}_i, \theta) \quad \text{model}$$

Error:

$$e_i = ||\mathbf{m}_i - \hat{\mathbf{m}}_i||$$

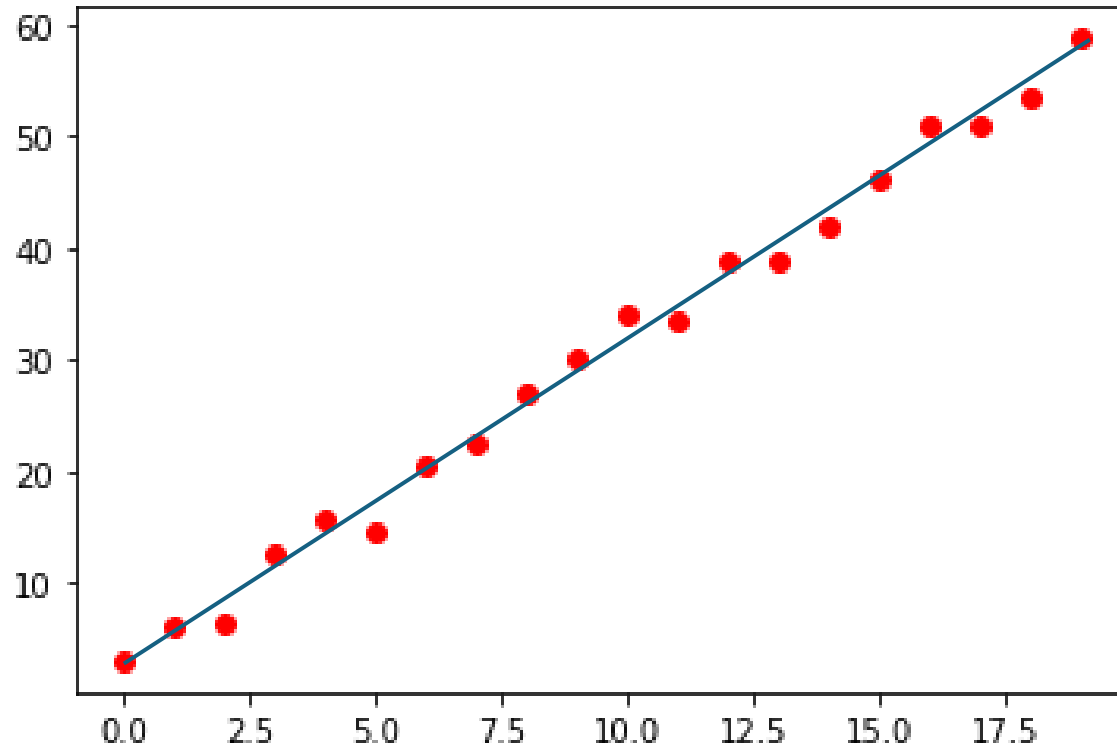
Objective Function to be minimized:

$$J(\theta) = \frac{1}{n} \sum_{i=1}^n ||\mathbf{m}_i - \hat{\mathbf{m}}_i||$$

Estimación de Parametros

[Example]

[Example: Estimation of a line]



MODEL

$$\ell \cdot \mathbf{m} = 0$$

$$\ell = [a \ b \ c]^T$$

$$\mathbf{m} = [x \ y \ 1]^T$$

$$ax + by + c = 0$$

(for each point: 1 equation)

$$\begin{array}{c} \mathbf{X} \\ \left[\begin{array}{ccc} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ & \vdots & \\ x_n & y_n & 1 \end{array} \right] \end{array} \begin{array}{c} \mathbf{a} \\ \left[\begin{array}{c} a \\ b \\ c \end{array} \right] \end{array} = \begin{array}{c} \left[\begin{array}{c} 0 \\ 0 \\ \vdots \\ 0 \end{array} \right] \end{array}$$

$$\mathbf{X}\mathbf{a} = \mathbf{0}$$

[Solution] $\mathbf{X}\mathbf{a} = \mathbf{0}$

$$\begin{bmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1,m-1} & x_{1m} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2,m-1} & x_{2m} \\ & & & \vdots & & \\ x_{n1} & x_{n2} & x_{n3} & \cdots & x_{n,m-1} & x_{n,m} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{m-1} \\ a_m \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

[Solution I] $\mathbf{X}\mathbf{a} = \mathbf{0}$

$(a_m = 1)$

$$\begin{bmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1,m-1} & x_{1m} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2,m-1} & x_{2m} \\ & & & \vdots & & \\ x_{n1} & x_{n2} & x_{n3} & \cdots & x_{n,m-1} & x_{n,m} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{m-1} \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

[Solution I] $\mathbf{X}\mathbf{a} = \mathbf{0}$

$(a_m = 1)$

$$\begin{bmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1,m-1} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2,m-1} \\ & & & \vdots & \\ x_{n1} & x_{n2} & x_{n3} & \cdots & x_{n,m-1} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{m-1} \end{bmatrix} = \begin{bmatrix} -x_{1m} \\ -x_{2m} \\ \vdots \\ -x_{nm} \end{bmatrix}$$

[Solution I] $\mathbf{X}\mathbf{a} = \mathbf{0}$

$$\overbrace{\begin{bmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1,m-1} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2,m-1} \\ & & & \vdots & \\ x_{n1} & x_{n2} & x_{n3} & \cdots & x_{n,m-1} \end{bmatrix}}^{\mathbf{Q}} \overbrace{\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{m-1} \end{bmatrix}}^{\mathbf{a}'} = \overbrace{\begin{bmatrix} -x_{1m} \\ -x_{2m} \\ \vdots \\ -x_{nm} \end{bmatrix}}^{\mathbf{r}}$$

$$\hat{\mathbf{a}}' = [\mathbf{Q}^T \mathbf{Q}]^{-1} \mathbf{Q}^T \mathbf{r} \quad \hat{\mathbf{a}} = \begin{bmatrix} \hat{\mathbf{a}}' \\ 1 \end{bmatrix}$$

[Solution 2] $\mathbf{X}\mathbf{a} = \mathbf{0}$
($\|\mathbf{a}\| = 1$)

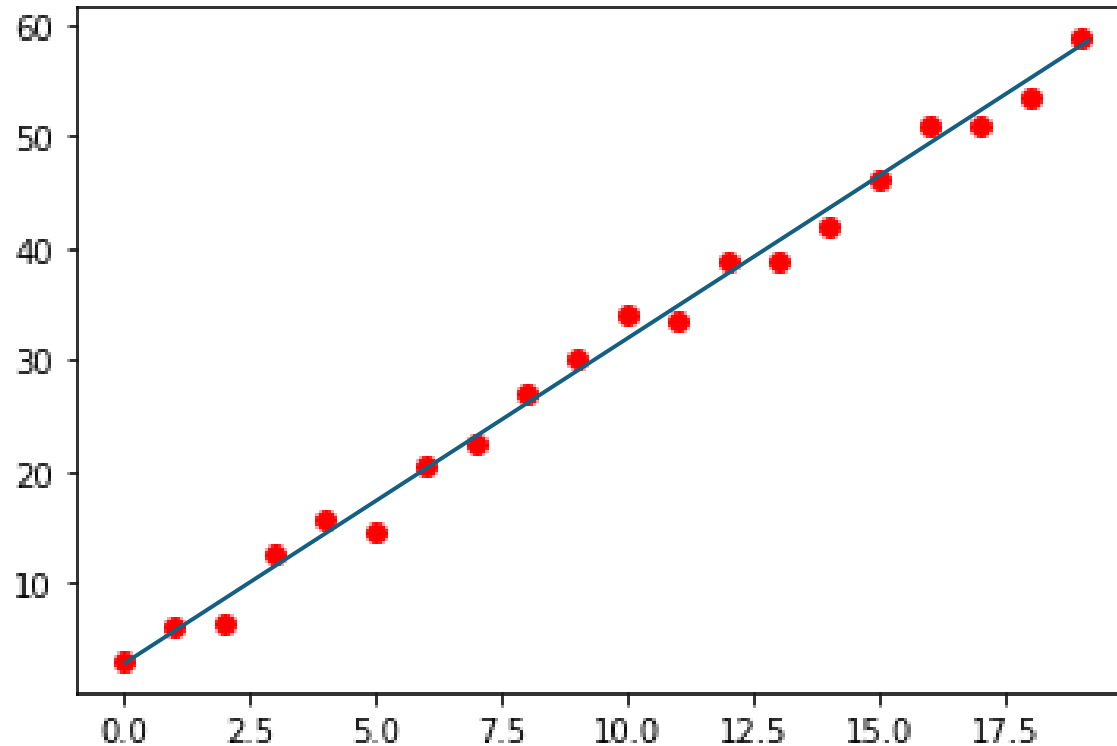
$$[\mathbf{U}, \mathbf{S}, \mathbf{V}] = \text{SVD}(\mathbf{X})$$

$$\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$$

$$\hat{\mathbf{a}} = \mathbf{V}_m \text{ (last column of } \mathbf{V}\text{)}$$

```
[U,S,V] = np.linalg.svd(X)
a_hat    = V[-1,:]
```

[Example: Estimation of a line]



MODEL

$$\ell \cdot \mathbf{m} = 0$$

$$\ell = [a \ b \ c]^T$$

$$\mathbf{m} = [x \ y \ 1]^T$$

$$ax + by + c = 0$$

(for each point: 1 equation)

$$\begin{array}{c} \mathbf{X} \\ \left[\begin{array}{ccc} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ & \vdots & \\ x_n & y_n & 1 \end{array} \right] \end{array} \begin{array}{c} \mathbf{a} \\ \left[\begin{array}{c} a \\ b \\ c \end{array} \right] \end{array} = \begin{array}{c} \left[\begin{array}{c} 0 \\ 0 \\ \vdots \\ 0 \end{array} \right] \end{array}$$

SOLUTION

$$\mathbf{X}\mathbf{a} = \mathbf{0} \rightarrow \hat{\mathbf{a}} = f(\mathbf{X})$$

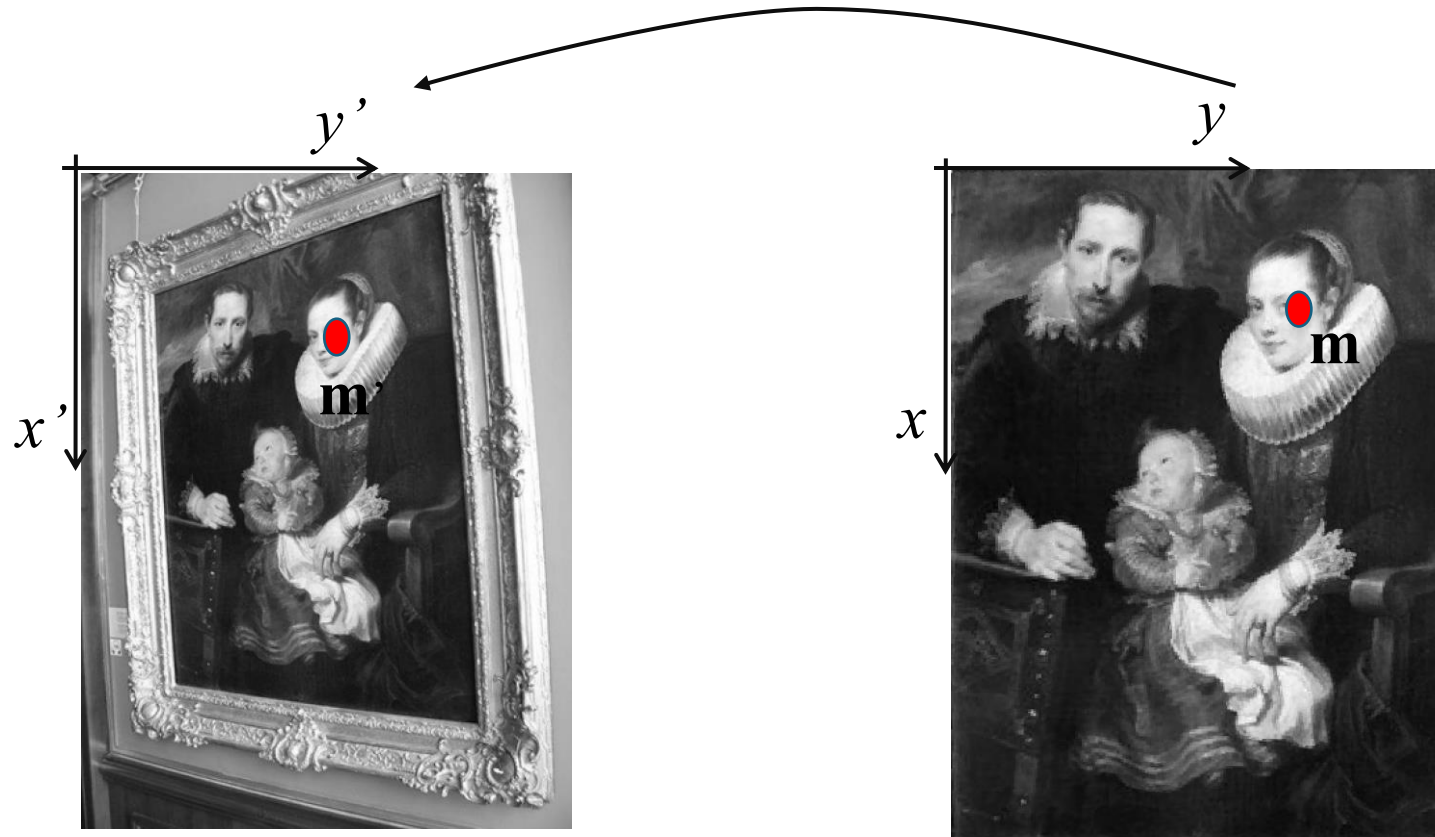
[In Computer Vision]

- Homography
- Camera Calibration
- Fundamental Matrix

[In Computer Vision]

- Homography
- Camera Calibration
- Fundamental Matrix

[Homography]



MODEL

$$\lambda m' = Hm$$

$$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

[Homography]

$$\lambda \begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

CORRESPONDENCE $(x', y') \leftrightarrow (x, y)$ (for each correspondence: 2 equations)

$$\begin{bmatrix} x & y & 1 & 0 & 0 & 0 & -x'x & -x'y & -x' \\ 0 & 0 & 0 & x & y & 1 & -y'x & -y'y & -y' \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ : \\ h_{33} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

[Homography]

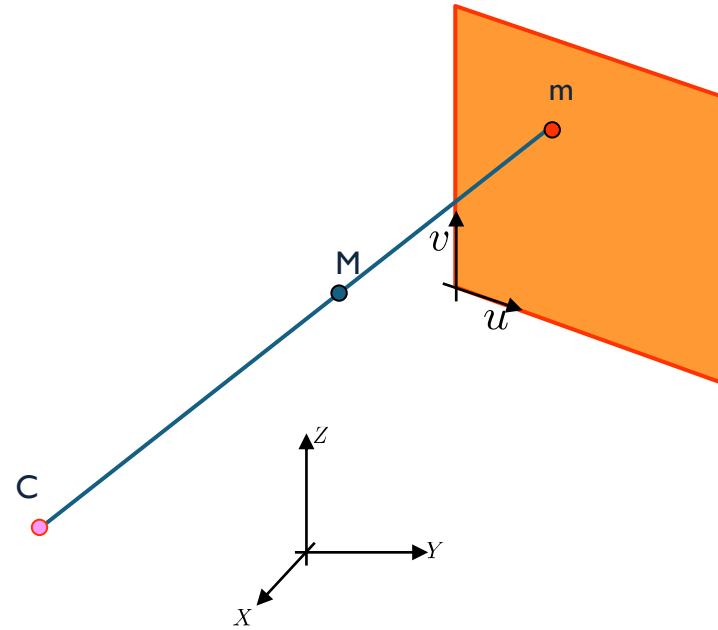
$$\begin{array}{c} \mathbf{X} \end{array} \begin{array}{c} \mathbf{a} \end{array} = \begin{array}{c} \begin{bmatrix} x_1 & y_1 & 1 & 0 & 0 & 0 & -x'_1 x_1 & -x'_1 y_1 & -x'_1 \\ 0 & 0 & 0 & x_1 & y_1 & 1 & -y'_1 x_1 & -y'_1 y_1 & -y'_1 \\ x_2 & y_2 & 1 & 0 & 0 & 0 & -x'_2 x_2 & -x'_2 y_2 & -x'_2 \\ 0 & 0 & 0 & x_2 & y_2 & 1 & -y'_2 x_2 & -y'_2 y_2 & -y'_2 \\ & & & & & \vdots & & & \\ x_n & y_n & 1 & 0 & 0 & 0 & -x'_n x_n & -x'_n y_n & -x'_n \\ 0 & 0 & 0 & x_n & y_n & 1 & -y'_n x_n & -y'_n y_n & -y'_n \end{bmatrix} \begin{bmatrix} h_{11} \\ h_{12} \\ \vdots \\ h_{33} \end{bmatrix} \end{array} = \begin{array}{c} \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix} \end{array}$$

$$\mathbf{Xa} = \mathbf{0}$$

[In Computer Vision]

- Homography
- Camera Calibration
- Fundamental Matrix

[Camera Calibration]



MODEL

$$\lambda \mathbf{w} = \mathbf{P} \mathbf{M} \quad \lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

[Camera Calibration]

$$\lambda \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} & p_{13} & p_{14} \\ p_{21} & p_{22} & p_{23} & p_{24} \\ p_{31} & p_{32} & p_{33} & p_{34} \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

CORRESPONDENCE $(u, v) \leftrightarrow (X, Y, Z)$ (for each correspondence: 2 equations)

$$\begin{bmatrix} X & Y & Z & 1 & 0 & 0 & 0 & 0 & -Xu & -Yu & -Zu & -u \\ 0 & 0 & 0 & 0 & X & Y & Z & 1 & -Xv & -Yv & -Zv & -v \end{bmatrix} \begin{bmatrix} p_{11} \\ p_{12} \\ \vdots \\ p_{34} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

[Solution I] $\mathbf{X}\mathbf{a} = \mathbf{0}$

$(a_m = 1)$

$$\begin{bmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1,m-1} & x_{1m} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2,m-1} & x_{2m} \\ & & & \vdots & & \\ x_{n1} & x_{n2} & x_{n3} & \cdots & x_{n,m-1} & x_{n,m} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{m-1} \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

[Solution I] $\mathbf{Xa} = \mathbf{0}$
 $(a_m = 1)$

$$\begin{bmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1,m-1} & x_{1m} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2,m-1} & x_{2m} \\ & & & \vdots & & \\ x_{n1} & x_{n2} & x_{n3} & \cdots & x_{n,m-1} & x_{nm} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{m-1} \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

```
def calibra(M,w):
    n = M.shape[1]

    for i in range(n):
        X = M[0,i]
        Y = M[1,i]
        Z = M[2,i]
        u = w[0,i]
        v = w[1,i]
        x1 = np.array([X,Y,Z,1,0,0,0,0,-u*X,-u*Y,-u*Z])
        x2 = np.array([0,0,0,0,X,Y,Z,1,-v*X,-v*Y,-v*Z])
        XXi = np.vstack([x1,x2])
        bbi = w[0:2,i].reshape(2,1)
        if i==0:
            XX = XXi
            bb = bbi
        else:
            XX = np.vstack([XX,XXi])
            bb = np.vstack([bb,bbi])
    a = np.linalg.lstsq(XX, bb, rcond=None)[0]

    # Matriz de proyección estimada
    P = np.array([
        [a[0], a[1], a[2], a[3]],
        [a[4], a[5], a[6], a[7]],
        [a[8], a[9], a[10], 1]
    ])
    return P
```


[Solution I] $\mathbf{X}\mathbf{a} = \mathbf{0}$

$(a_m = 1)$

$$\begin{bmatrix} x_{11} & x_{12} & x_{13} & \cdots & x_{1,m-1} \\ x_{21} & x_{22} & x_{23} & \cdots & x_{2,m-1} \\ & & & \vdots & \\ x_{n1} & x_{n2} & x_{n3} & \cdots & x_{n,m-1} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_{m-1} \end{bmatrix} = \begin{bmatrix} -x_{1m} \\ -x_{2m} \\ \vdots \\ -x_{nm} \end{bmatrix}$$

[Camera Calibration]

$$\begin{array}{c}
 \mathbf{X} \qquad \qquad \qquad \mathbf{a} \\
 \underbrace{\hspace{15em}} \quad \underbrace{\hspace{2em}}
 \end{array}$$

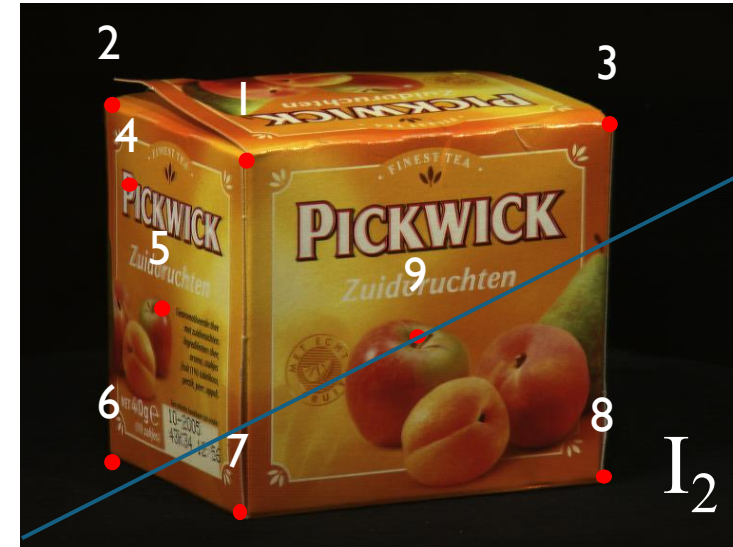
$$\begin{bmatrix}
 X_1 & Y_1 & Z_1 & 1 & 0 & 0 & 0 & 0 & -X_1 u_1 & -Y_1 u_1 & -Z_1 u_1 & -u_1 \\
 0 & 0 & 0 & 0 & X_1 & Y_1 & Z_1 & 1 & -X_1 v_1 & -Y_1 v_1 & -Z_1 v_1 & -v_1 \\
 X_2 & Y_2 & Z_2 & 1 & 0 & 0 & 0 & 0 & -X_2 u_2 & -Y_2 u_2 & -Z_2 u_2 & -u_2 \\
 0 & 0 & 0 & 0 & X_2 & Y_2 & Z_2 & 1 & -X_2 v_2 & -Y_2 v_2 & -Z_2 v_2 & -v_2 \\
 & & & & & & \vdots & & & & & \\
 X_n & Y_n & Z_n & 1 & 0 & 0 & 0 & 0 & -X_n u_n & -Y_n u_n & -Z_n u_n & -u_n \\
 0 & 0 & 0 & 0 & X_n & Y_n & Z_n & 1 & -X_n v_n & -Y_n v_n & -Z_n v_n & -v_n
 \end{bmatrix}
 \begin{bmatrix}
 p_{11} \\
 p_{12} \\
 \vdots \\
 p_{34}
 \end{bmatrix}
 =
 \begin{bmatrix}
 0 \\
 0 \\
 0 \\
 0 \\
 \vdots 0 \\
 0
 \end{bmatrix}$$

$$\mathbf{Xa} = \mathbf{0}$$

[In Computer Vision]

- Homography
- Camera Calibration
- Fundamental Matrix

[Fundamental Matrix: Epipolar Geometry]



Epipolar
Line

MODEL

$$\mathbf{m}_2^T \mathbf{F} \mathbf{m}_1 = 0 \quad \begin{bmatrix} x_2 & y_2 & 1 \end{bmatrix} \begin{bmatrix} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} = 0$$

[Fundamental Matrix: Epipolar Geometry]

$$\begin{bmatrix} x_2 & y_2 & 1 \end{bmatrix} \begin{bmatrix} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} = 0$$

CORRESPONDENCE $(x_1, y_1) \leftrightarrow (x_2, y_2)$ (for each correspondence: 1 equation)

$$\begin{bmatrix} x_1 x_2 & y_1 x_2 & x_2 & x_1 y_2 & y_1 y_2 & y_2 & x_1 & y_1 & 1 \end{bmatrix} \begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{bmatrix} = 0$$

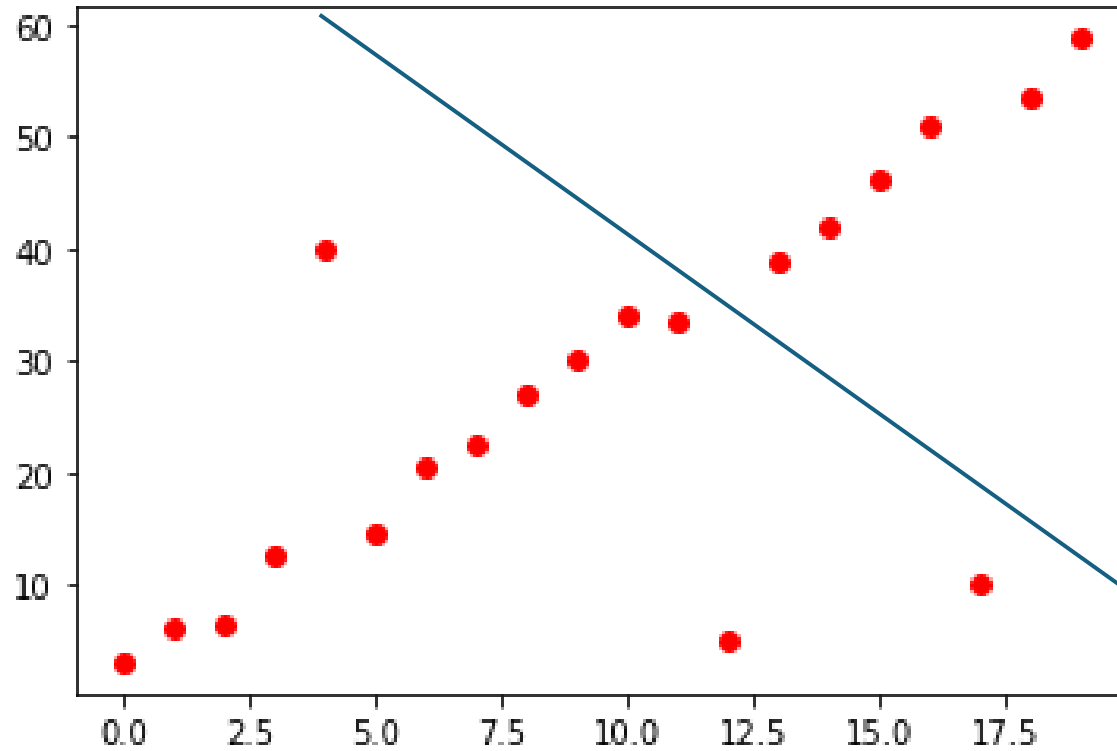
[Fundamental Matrix: Epipolar Geometry]

$$\begin{array}{c} \mathbf{X} \end{array} \begin{array}{c} \mathbf{a} \end{array} = \begin{array}{c} \begin{bmatrix} x_1^1 x_2^1 & y_1^1 x_2^1 & x_2^1 & x_1^1 y_2^1 & y_1^1 y_2^1 & y_2^1 & x_1^1 & y_1^1 & 1 \\ x_1^2 x_2^2 & y_1^2 x_2^2 & x_2^2 & x_1^2 y_2^2 & y_1^2 y_2^2 & y_2^2 & x_1^2 & y_1^2 & 1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ x_1^n x_2^n & y_1^n x_2^n & x_2^n & x_1^n y_2^n & y_1^n y_2^n & y_2^n & x_1^n & y_1^n & 1 \end{bmatrix} \begin{bmatrix} F_{11} \\ F_{12} \\ F_{13} \\ F_{21} \\ F_{22} \\ F_{23} \\ F_{31} \\ F_{32} \\ F_{33} \end{bmatrix} \end{array} = \begin{array}{c} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \end{array}$$

$$\mathbf{X}\mathbf{a} = \mathbf{0}$$

[Outliers]

[Example: Estimation of a line - OUTLIERS]



MODEL

$$\ell \cdot \mathbf{m} = 0$$

$$\ell = [a \ b \ c]^T$$

$$\mathbf{m} = [x \ y \ 1]^T$$

$$ax + by + c = 0$$

(for each point: 1 equation)

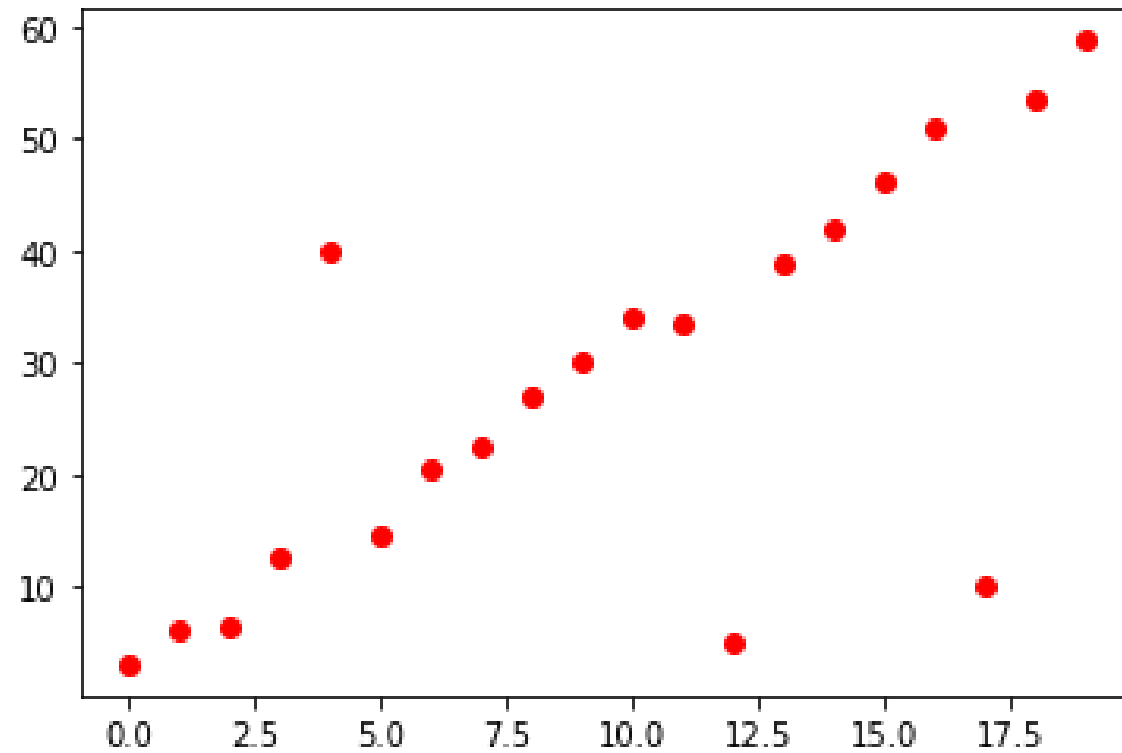
$$\begin{matrix} \mathbf{X} & \mathbf{a} \\ \left[\begin{array}{ccc} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ & \vdots & \\ x_n & y_n & 1 \end{array} \right] & \left[\begin{array}{c} a \\ b \\ c \end{array} \right] \end{matrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

SOLUTION

$$\mathbf{X}\mathbf{a} = \mathbf{0} \rightarrow \hat{\mathbf{a}} = f(\mathbf{X})$$

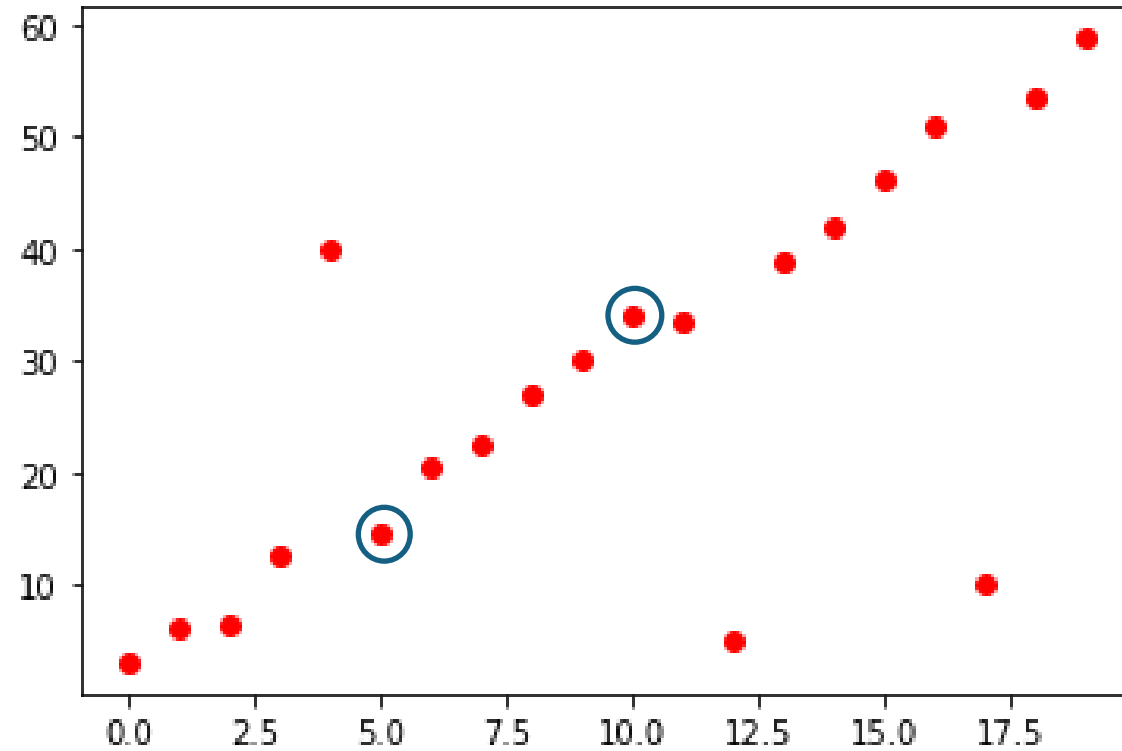
[How to remove the OUTLIERS]

RANSAC



[How to remove the OUTLIERS]

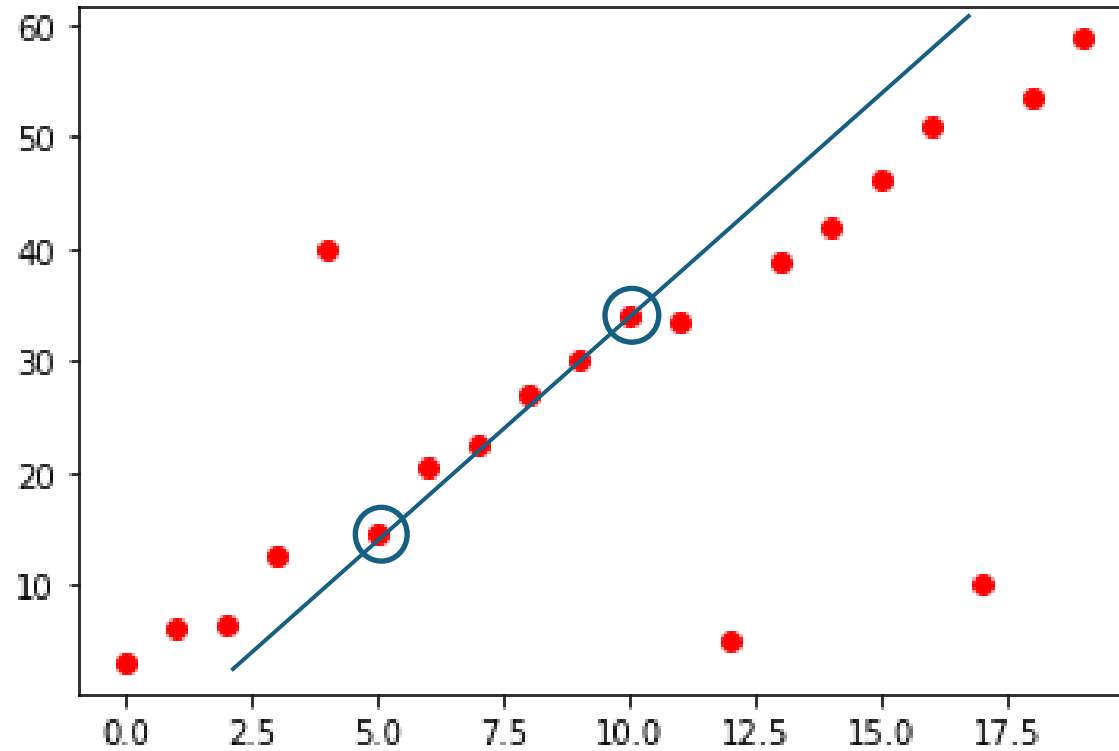
RANSAC



I. Select random 2 points

[How to remove the OUTLIERS]

RANSAC

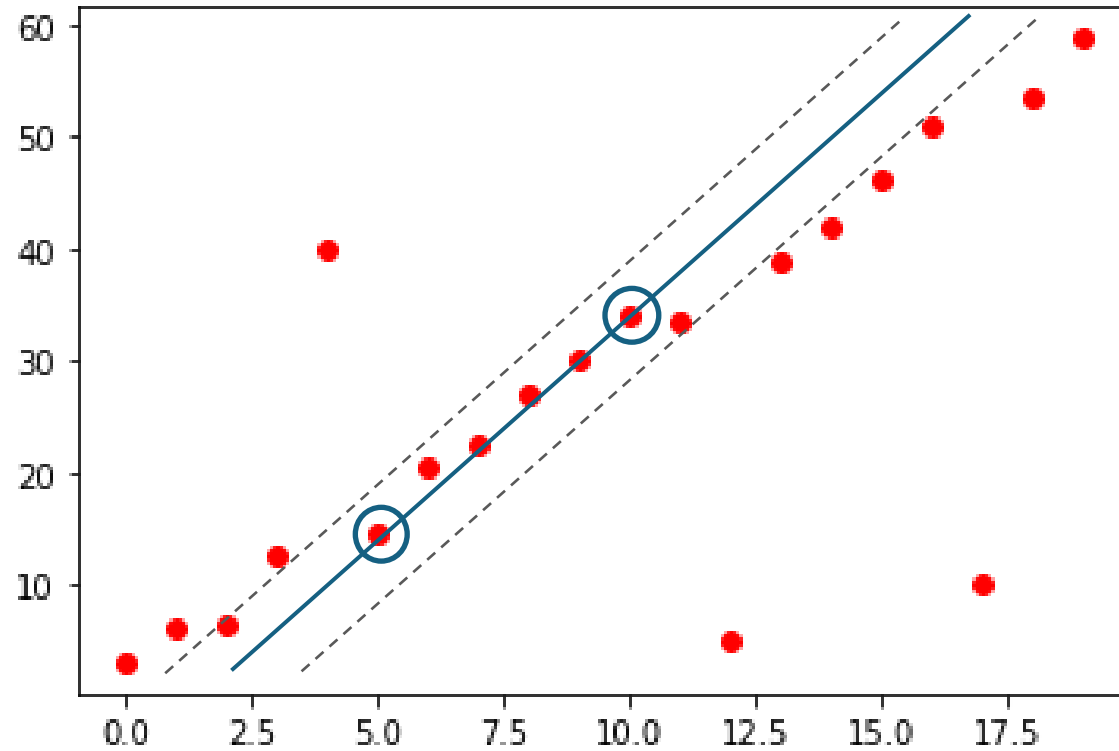


1. Select random 2 points

2. Estimate the line

[How to remove the OUTLIERS]

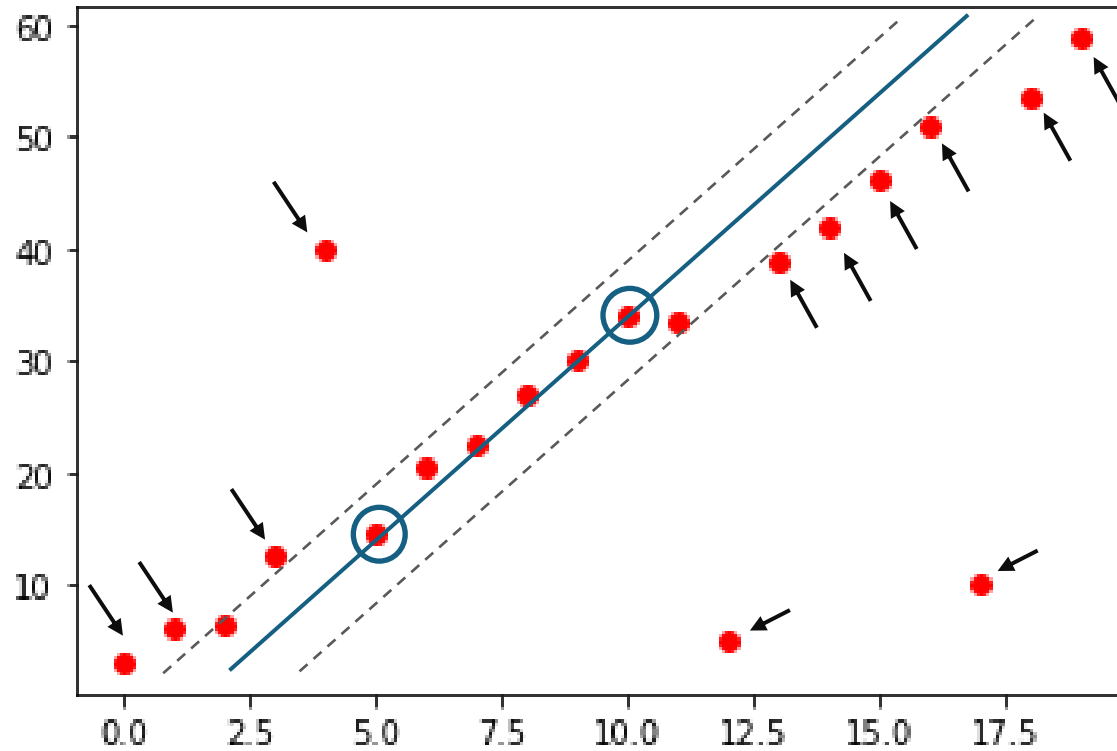
RANSAC



1. Select random 2 points
2. Estimate the line
3. Define a tolerance range

[How to remove the OUTLIERS]

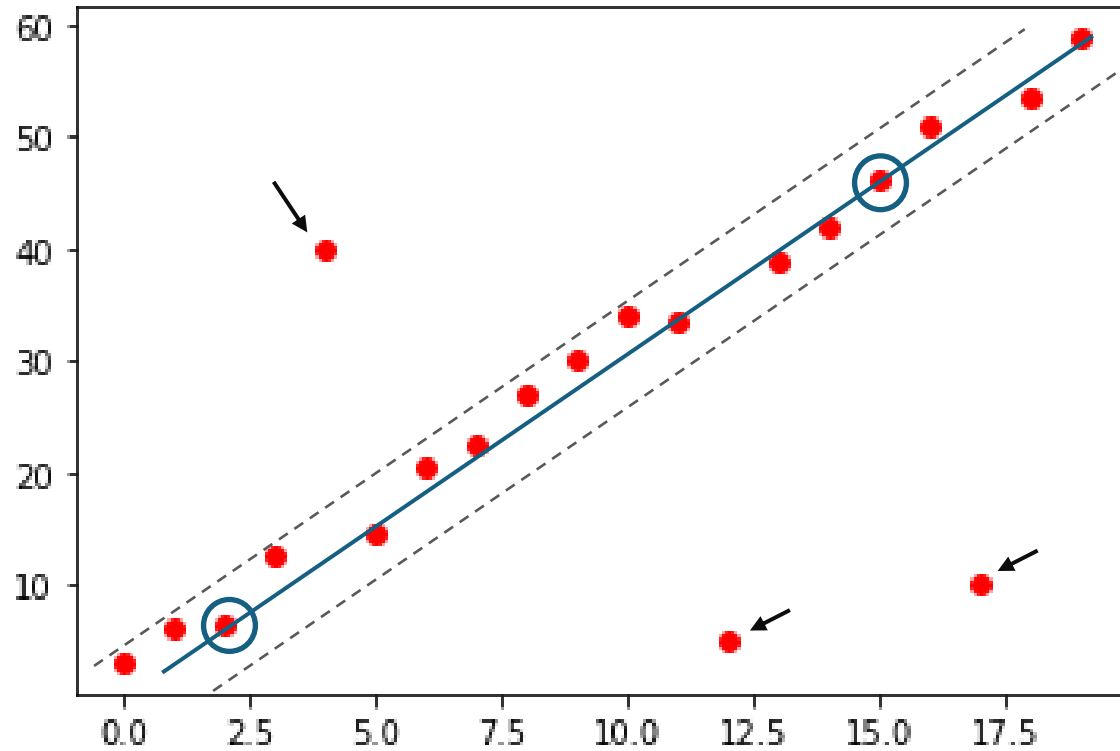
RANSAC



1. Select random 2 points
2. Estimate the line
3. Define a tolerance range
4. Count the outliers

[How to remove the OUTLIERS]

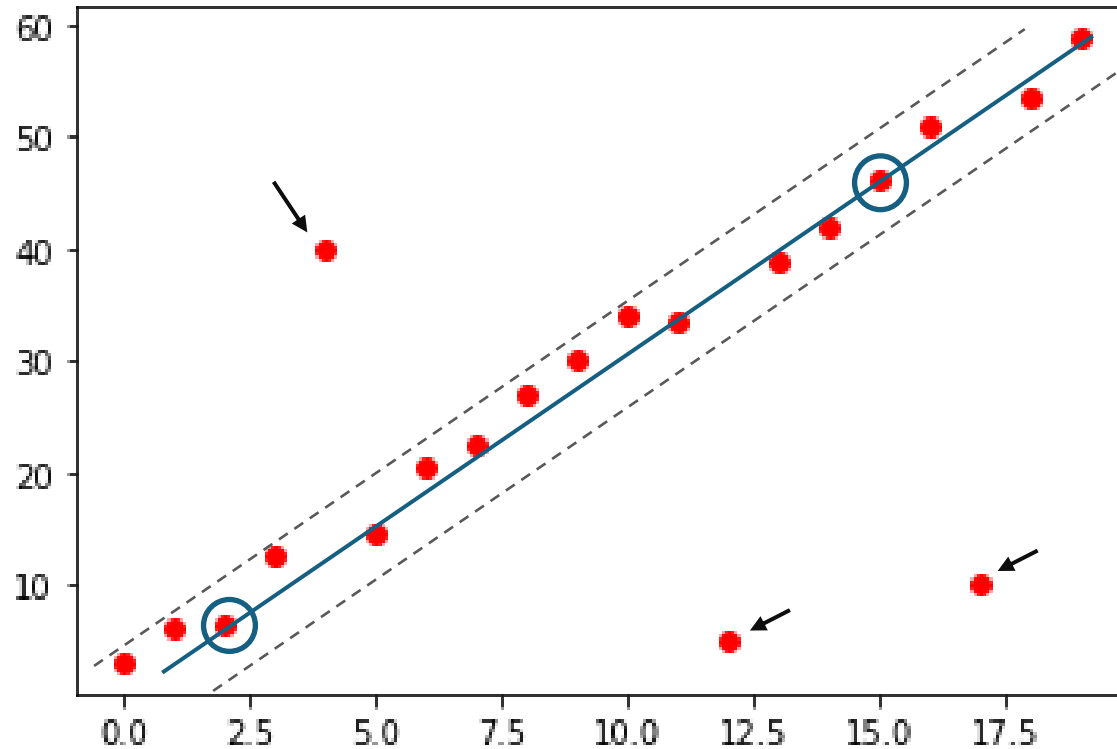
RANSAC



1. Select random 2 points
2. Estimate the line
3. Define a tolerance range
4. Count the outliers
5. Repeat N times and select the line that minimizes the number of outliers

[How to remove the OUTLIERS]

RANSAC



1. Select random 2 points(*)
2. Estimate the line
3. Define a tolerance range
4. Count the outliers
5. Repeat N times and select the line that minimizes the number of outliers

(*) Minimal number of points to estimate the parameters of the model

Corner detection (Chess)

[In Computer Vision]

- Homography
- Camera Calibration
- Fundamental Matrix

SIFT matching points

Corresponding points can be found automatically...

Outliers can be removed using RANSAC